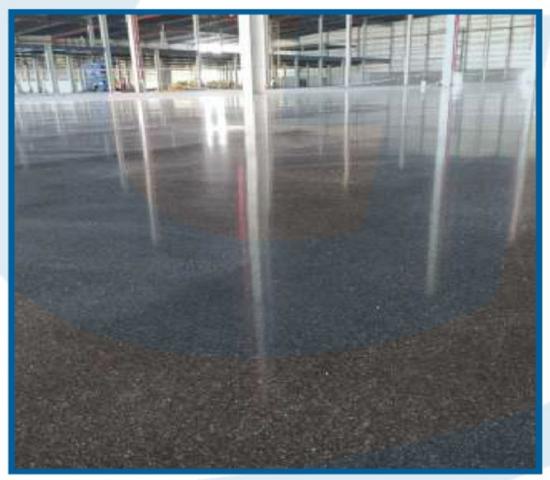

LASER SCREED TECHNOLOGY

FLOORING SERVICES

Contact :

GANGADHAR CONCRETE SOLUTION

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ABOUT US: GANGADHAR CONCRETE SOLUTION

is the first laser screed concrete flooring company based in **Jaipur Rajasthan.**
Started by young entrepreneur who wanted to match high level professionalism and quality standard in concrete flooring. We come with value engineered flooring solutions and a level of assurance that goes way beyond industry bonds.
We are the fastest growing flooring solution providers based on international codes like TR34 4th edition from UK concrete society, American codes (ASTM E 1155) and DIN codes meeting today's surface regularity requirements for free movement and defined movement floors in warehousing and industrial sector. We continuously invest in ongoing R&Ds to keep ourselves well-conversant with the latest technological developments.
We are having expertise in doing concrete flooring works for industry, warehouse, parking, stockyard etc. by using latest technology machinery. We have well trained and qualified team of engineers and workers to achieve high quality standard work.

VISION:

We believe in going above our client's expectation to deliver outstanding services every time by maintaining long term and healthy relations by providing best quality concrete flooring work .

MISSION:

To become one of the most reliable and trustworthy concrete flooring company in pan India by providing best quality of work on time.

MOTTO:

Your Floor, Our Expertise.

OUR SERVICES

FM1/ FM2 Grade -
Laser Screed Floors

VNA Floors -
DM1/DM2 Category

Steel Fiber Reinforced
Concrete Flooring

Jointless Floors

Concrete Polishing
and Densification

Epoxy Flooring

OUR PROJECT

Plasli Weave Industries LLP, Udaipur

Renew Power Dholera, Ahmedabad

Kuber India Mart, Jaipur

Keracoll India, Jaipur

PLATE DOWEL & SLEEVE

Ideal for load transfer between concrete slabs

How Plate Dowel & Sleeve works?

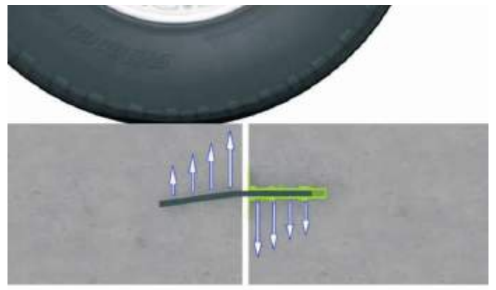
When there is a movement across the joint such as forklift, MHE, and other machinery that joints between the concrete slab, it needs to be strengthened since there is load transfer between adjacent slabs, and these joints should be capable of bearing that load which is acting on that joint and also needs to transfer it from one slab to another without disturbing the serviceability of the slab.

This could be achieved by introducing plate dowels (Diamond or Rectangular) which are capable of increasing the shear, bending strength and overall load capacity of the joint.

The effective loads at joints are reduced by load transfer through Plate dowels and protects from the failure of the construction joint. Duraplate conforms to the EN 10025-2:2004, material is ABS and Design refs. to TR-34.

Rectangular Sleeve With Plate Dowel

Diamond Sleeve With Plate Dowel



Load Transfer Mechanism

ADVANTAGES

- Provides load transfer across the joint
- Increased bending strength and shear strength of joint
- Enhanced load-bearing capacity
- Minimises differential deflection between adjacent slabs
- Allows diagonal shrinkage movement in the horizontal plane
- Allow perpendicular and parallel differential shrinkages
- Maximizes surface area of joint line
- Eliminates drilling or processing on formworks
- Reduces risk of restraint
- Speed and accuracy of the dowel placement

WHY US?

We understand the importance of correctly designing and installing floors, often referred as super flat floor.

We provide turnkey solution with single source responsibility of project starting from inspection to completion, covering design, engineering, detailing, constructing and certification.

We use latest designs and detailed softwares which provide robust solution. Latest technology machinery is used with well-trained operators for accuracy of work.

Speedy and planned execution drastically cut down time and costs of project.

We provide live update to client of their projects.

We offer a wide range of services that deliver floor construction according to the customer's needs.

We provide pre and post service to client with same efficiency and interest.

KEY SERVICES:

Laser screed flooring (FM1/FM2 Grade).
Applicable / Suitable for the areas where materials handling equipment operates in random, non-defined directions and have an infinite number of travel paths.

VNA flooring (DM1/DM2)
Applicable / Suitable where very narrow aisles MHE are employed for stacking of material at 12 to 13 mtr height.

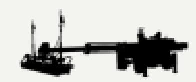
Steel fiber reinforced concrete flooring.
Applicable / Suitable where design for the high load bearing capacity area, which decrease cracking and increase durability of floor.

Jointless floor.
Applicable / Suitable for the smooth operation which has very few joints for high load bearing and low maintenance.

Concrete polishing and densification
Applicable / Suitable for the dust proof, maintenance free, slip resistance, ambient lighting and long lasting of floor.

Testing and certification.
Conducting tests of floor by third party-chartered engineer and also certify your floor for FM/DM category as per TR34 code.

LASER SCREED FLOORS |



(FM1/FM2 Grade)

Like most industries, constructing concrete floors is competitive and demanding. Owners want their floors done faster and flatter than ever before. The drive to achieve these goals fostered an eagerness among concrete contractors to push the performance envelope. The laser screeding technique allows you to lay concrete floors in wider bays of any size.

It dramatically reduces the time, reduces the number of construction joints, improves the flatness of floors. It also reduces the number of labor on the site. It eliminates the need of fixing guide rails to every four meters. Laser Screed simultaneously cuts, vibrates & levels the concrete in a single pass. All above operations are controlled by laser transmitters, receivers & computers. So manual errors are eliminated.

Using the latest laser screed machinery, we can lay and high tolerance concrete floor slab of up to 1200m2 in a single day. The tolerances achieved with this machine in free movement slabs are FM1, FM2 (special), FM2, and FM3 in accordance with the TR34 Document.

Floor Class	Floor Classification For Free Movement	Property E Levelness	Property F Flatness
FM1	Where every high standard of flatness and levelness are required. Reach trucks operating at above 13m without side-shift	4.5	1.8
FM2	Reach trucks operating at 8 to 13m without side shift	6.5	2.0
FM3	Retail floors to take directly applied finishes Retail trucks operating at up to 8m without side-shift Retail trucks operating at up to 13m without side-shift	8.0	2.2
FM4	Retail floors to take applied screeds. Workshops and manufacturing facilities where MHE lift heights are restricted to 4m	10.0	2.4

Note: Side-shift is the ability of a truck to adjust the pallet transversely to the fork direction

ADVANTAGES |

- Assures laser-precise flatness and
- VNA floors are best suited for ASRS
- Due to fewer joints get maximum
- Aisle space has no joints enabling
- Achieve more precise flatness and
- Reduces the overall construction




ARMOUR JOINT

For protecting the heavy duty operational / construction joint

How Armour Joints work

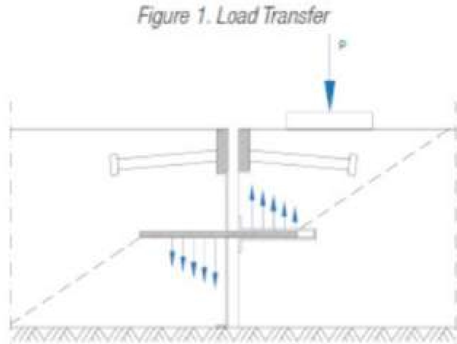
Armour joints are installed into position at designed construction joint, once the concrete is placed the shrinkage forces are generated by the drying concrete slabs at the time of curing process.
This leads to the shearing of the plastic bolts which allows the two steel profiles to open at a permitted joint opening.

Permits the minor controlled slab movements, caused by the drying shrinkage and thermal variation in horizontal direction of the slab planes and restrict vertical displacement of the slabs. Armour Joint conforms to joint aris EN 10277-1:2018, shear studs conform to EN ISO 13918 :2017, Duraplate conforms to EN 10025-2:2004 and material is ABS and Design refs. to TR-34.

Applications of Armour Joints

- **Industrial Floors**
- **Warehouses and Distribution Centres**
- **Industrial Storage Facilities**
- **Roads and Pavements**
- **Airports and Runways**
- **Dockyards**

Figure 1. Load Transfer



ADVANTAGES

- Improves bearing strength of joint in terms of shear and bending
- Resistant to twisting caused by the impact of wheel ed traffic
- Efficiently transfers load between adjacent concrete slabs with plate dowels.
- Allows controlled horizontal slab movement, preventing random cracking.
- Eliminate vertical movement between slabs during concrete contraction, for a level floor surface.
- Mitigates joint repairs, curling and spalling
- Allows for continuous pouring on both sides of the joint
- Armours heavy duty joints

MACHINERY:

Laser screed machine (somero S-158C)
Ride on Trowel (Husqvarna CRT36)
Walk behind Trowel (Husqvarna MCT 36-5)
Groove cutting machine
Hardener spreader
Bull float
Bump cutter
Laser leveller
Auto leveller








PROCESS:

Pre survey:
Sub base is the most crucial part for the life span of floor. Thus, our Engineers visits the site and carry outs level survey along with plate bearing test (1. Test at every 2000 m & 2. Of floor). Which will ensure that the sub base is prepared to the correct tolerance and is compact properly with proper level.

Pre pouring:

- Slip membrane installation (200 microns LDPE sheet)

To reduce the friction between slab and sub base and to reduce the restraint to drying and shrinkage which reduce the risk of unplanned cracking.

- Form work and joint

Use adjustable timber formwork with "L" angle. Steel armoured joint are used for heavy duty load transfer system where ever construction joints fall in trafficked areas. Diamond plate dowels of 6mm thick (100x100mm @450mmc/c) is fixed for every construction joint to transfer load.

- Steel laying and Isolation joint wall & column

Locally available steel laying as per design. Fixing 10mm thick compressible material for isolation joint between wall and slab, 20mm thick between column and slab.

Slab Pouring:
At the time of pouring checking slump and temperature of concrete. Pouring of concrete laying and compacting by somero laser screed machine for quality and flatness. After laying and levelling non-metallic floor hardener is mechanically applied (3.5-4kg/sq. metre) by semi mechanized topping spreader to improve abrasion resistance.

Flatness and Finishing:
Experience finishing team do the finishing of slab by ROT and Power trowel.

Comparison

MANUAL SCREEDING	LASER SCREEDING
• Relatively slower than laser screed. • It requires more number of joints on floor. • Chances of Human error are more and thus quality standards are compromised. • No certification as such.	• Increased productivity and efficiencies. • Reduces number of construction joints and provides more durability and strength. • Laser precise flatness and levelness every time. • FM Certification for floor. •

Benefits:

To Owner:
Lower long term operational costs related to repair & maintenance.
The building becomes more valuable.
Shorter construction period & speeds up building revenue generation.

To End User:
Lower use-costs & operational disruption.
Lower damaged goods costs.
Higher productivity/ Increases Outputs.

High quality floor = Low cost solution